INCH POUND

MIL-C-3098/98E
5 May 1993
SUPERSEDING
MIL-C-3098/98D
8 March 1983

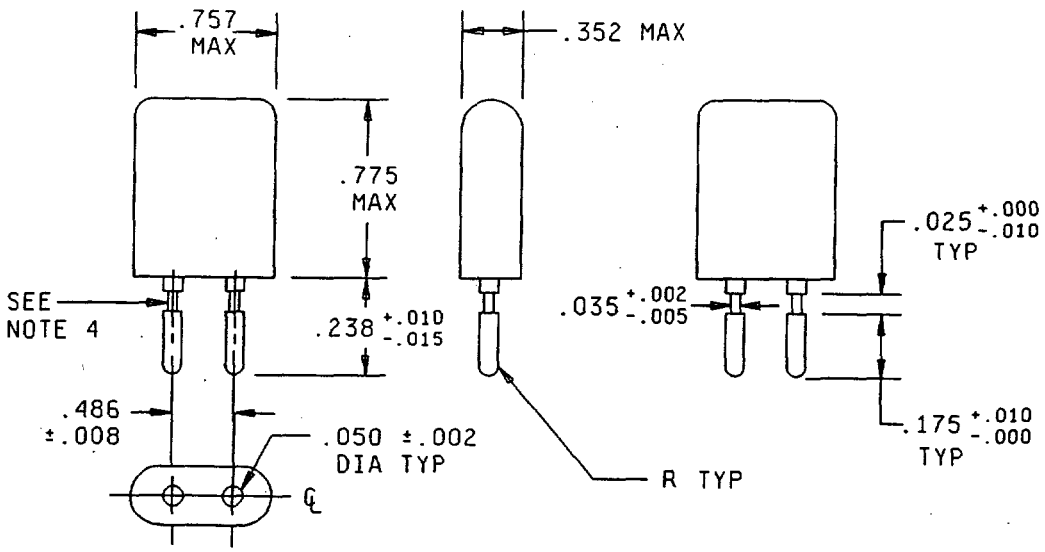
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR121/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 2 to 10 MHz; fundamental; controlled; antiresonance.



Inches	mm	Inches	mm
.002	.05	.050	1.27
.005	.13	.175	4.45
.008	.20	.238	6.05
.010	.25	.352	8.94
.015	.38	.486	12.34
.025	.64	.757	19.23
.035	.89	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR121/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-27/U (MIL-H-10056/14).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2 to 10 MHz, inclusive.

Capacitance, shunt: 6 pF, maximum.

Frequency tolerance: ±5 ppm at +85°C ±0.5°C.

Antiresonance, load capacitance: 50.0 ±0.5 pF.

Mode of oscillation: Fundamental.

Temperature range:

Operating: +85°C ±5°C.

Operable: -55°C to +90°C, inclusive.

Drive level, rated: 200 µW.

Crystal quality (Q): 300,000 minimum. 1/

Temperature coefficient of parallel resonant frequency: 0.2 ppm/°C at +85°C maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Aging:

Frequency change permitted: ±0.4 ppm per week for the last 2 weeks of a 4-week aging test.

1/ $Q = \frac{1}{4\pi(\Delta f)(C_T)(R)}$

Where: Δf = difference, in Hz, between the series and antiresonant frequency at C_L (load capacitance) of 50 pF.

C_T = Capacitance: C_o (shunt capacitance of the crystal unit) plus C_L (load capacitance) of 50 pF.

R = Motional-arm resistance of the crystal unit.

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QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTE: NATO guidance type NXT 36.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0678-13)

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